



Structure and Stability of the Rhombus Family of Relative Equilibria under General Homogeneous Forces

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Abstract

Let $N > 2$ and $n > 1$. Among the classes of symmetric relative equilibria of the N -body problem whose symmetry group is one of the dihedral groups D_n , the rhombus family stands out as the only noncollinear family for which the symmetry implies that the stability polynomial is fully factorizable. The present paper discusses basic properties and linear stability of the rhombus family assuming the forces of interaction depend on the mutual distances raised to an arbitrary real exponent $2a + 1$. In a suitable parameter plane, the family of rhombus relative equilibria forms a pencil of graphs which foliates the union of an open unit square and an open rectangle obtained from the unit square by a reflection and an inversion. We show that all rhombus relative equilibria are linearly stable if $a > -1$, that they are all unstable for a in the interval bound by $-4 - 2\sqrt{2} \approx -6.82$ and $4(\sqrt{3} - 2) \approx -1.07$, and that stability and instability depend on mass values for the remaining values of a . These results impose limitations on the validity of Moeckel's dominant mass stability conjecture in the context of generalized N -body problems.

Keywords N -body problem · Relative equilibria · Linear stability

Mathematics Subject Classification 70F10 · 37N05 · 70Fxx · 37Cxx

1 Introduction

Let N particles (or bodies) of positive masses $m_1, \dots, m_N$ be located at $\mathbf{q}_1, \dots, \mathbf{q}_N \in \mathbb{R}^d$. The particles are subject to mutual attraction in such a way that the j th particle exerts on the i th particle a force given by

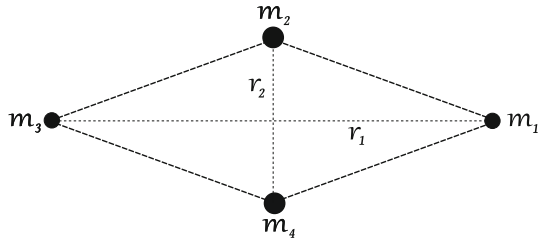
$$\mathbf{F}_{ij} = m_i m_j \|\mathbf{q}_i - \mathbf{q}_j\|^{2a} (\mathbf{q}_j - \mathbf{q}_i), \quad (1.1)$$

where a is a real number. A force given by the above expression is called a *generalized homogeneous force with exponent a* . For a fixed a , the problem of determining the dynamics

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Fig. 1 A rhombus configuration of four point masses



of the N particles interacting through mutual generalized homogeneous forces will be referred to as a *generalized N -body problem*. The classical Newtonian N -body problem is the case $a = -3/2$.

Briefly speaking, a relative equilibrium (abbreviated as RE) is a planar solution of the generalized N -body problem in which the bodies rotate uniformly about their center of mass. Notice the configuration formed by the particles moving in a relative equilibrium is preserved throughout their motion. Such a configuration is known as a *central configuration*. If $d = 2$, it is a common practice to use the terminologies “relative equilibrium” and “central configuration” interchangeably, although, depending on initial velocities, other (Keplerian) motions starting at a central configuration are possible. Relative equilibria of the Newtonian N -body problem is a traditional research topic whose origin goes back to the work of L. Euler on the three-body problem. For a given ordering of three point masses on a line, Euler proved that there corresponds a unique triple of positions forming a relative equilibrium. He also appears to have analyzed the stability of such solutions, being much later corrected by Liouville, cf. [1] and references therein. Reference [8] contains a brief overview of the history of linear stability analysis of relative equilibria in Celestial Mechanics.

Among the classes of symmetric N -body relative equilibria, $N > 2$, with symmetry group equal to one of the dihedral groups D_n , $n \geq 2$, the rhombus family stands out as the only family of noncollinear RE for which the stability matrix achieves maximum simplification, see Ref. [5], section 10, for a detailed explanation. As an algebraic feature of the equations of motion, this simplification is equally valid for all exponents a , thus allowing for a study of stability where a is arbitrary. The implementation of such a study depends on a detailed understanding of how the family is structured in terms of the masses at the vertices of the rhombus, the lengths of the diagonals, and the force exponents. Thus our primary goal is to clarify the structure determined by the interplay between masses, shape and exponents. We note that, in the context of the Newtonian four-body problem, the rhombus family has been studied in the past century by Brumberg [4] and Longley [6].

From the perspective of the generalized four-body problem, the rhombus family is most simply described as a two-parameter family of relative equilibria, one of the parameters being the exponent a , and the other, the length of one of the semidiagonals. Consider Fig. 1. The main theorem in Albouy et al. [3] imposes that the masses on the same diagonal must be equal; an specific argument for this fact is given in Sect. 2. From the homogeneity of the relative equilibrium equations, the rhombus family of relative equilibria is completely described by the subfamily of rhombi with $r_2 = 1$ and $m_2 = m_4 = 1$. If $r_1 = r$ and $m_1 = m_3 = m$, the RE equations allow us to express m as an exponential rational function of r and a . A key symmetry becomes noticeable. If $m < 1$ and $r > 1$ are the parameters of a (prolate) rhombus relative equilibrium for a given exponent a , the same fact is true for the (oblate) rhombus with $1/m$ and $1/r$. We exploit this *inversion symmetry* through a suitable Möbius transformation in Sect. 2 in order to determine the structure of the rhombus family. Using a and $s = r^2$

as parameters, we show that, up to the inversion symmetry, the family of rhombus relative equilibria can be viewed as a foliation of the rectangle $[1, 3) \times (0, 1]$ in the (s, m) -plane by graphs. These graphs form a pencil of curves having two focal points, namely the square with equal masses at its vertices, and a prolate rhombus with a pair of zero masses at the endpoints of its longer diagonal. The aforementioned Möbius transformation converts the inversion symmetry in the s parameter into a reflection symmetry in a new parameter v and, in terms of a and v , the expression for m at a rhombus RE, as well as other relevant expressions, achieve a form which allows for a more productive analysis.

The stability of the rhombus as a relative equilibrium in the Newtonian four-body problem is examined in [4], where the author argues that all rhombi are unstable since the discriminants of two of the four quadratic factors of the stability polynomial turn out to be negative. The other two quadratic factors are ignored since they originate from the action of continuous groups of rotations, translations and homotheties.

In Sect. 3 we provide a brief review of the basic theory of linear stability, followed by a proof that all relative equilibria in $\mathbb{R}^3$ are normally linearly stable. Our statement slightly strengthens proposition 2.1 of [8]. In “Appendix A”, we employ the technique of [5] to deduce the factorization of the stability polynomial of the rhombus into quadratics as an application of group representation theory [12]. This factorization is used in Sect. 3 to obtain explicit formulas for the eigenvalues of the rhombus and conditions for linear stability. Sections 4–6 contain a detailed study of the two quadratic factors that determine stability, and the stability results that follow therefrom. In either case, the key expression to be analyzed is the *normalized discriminant*, a factor of the actual discriminant which determines the sign thereof and is invariant under inversion. Next we highlight some of our stability results, which will be stated up to the inversion symmetry.

All rhombus relative equilibria are linearly stable when $a > -1$. When $a = -1$, two rhombi become unstable, one of which is the square with equal masses at its vertices. More precisely, the square becomes unstable with respect to displacements in a plane $\Pi^{(\phi, \psi)}$, while remaining linearly stable with respect to displacements in the complementary plane $\Pi^{(\tau, \alpha)}$ as long as a is greater than -1.03 . Here $\Pi^{(\tau, \alpha)}$ and $\Pi^{(\phi, \psi)}$ denote the two-dimensional subspaces of $\mathbb{R}^8$ which are invariant under the stability matrix and determine the quadratic polynomials mentioned in the previous paragraph, and the superscripts τ, α, ϕ, ψ designate the irreducible representations of the dihedral group D_2 . Moving along the a -axis in the negative direction, we have a short interval $\approx [-1.07, -1)$ where a neighborhood of the (most) oblate rhombus $\mathcal{F}$ with a pair of zero masses on its longest diagonal is linearly stable and a neighborhood of the square $\mathcal{Q}$ is unstable. The short interval of mixed stability is followed by an interval bounded approximately by -6.82 and -1.07 where all rhombus relative equilibria are unstable. A neighborhood of $\mathcal{Q}$ is actually linearly stable with respect to displacements in $\Pi^{(\tau, \alpha)}$ if $a \gtrsim -3.89$, and for $a \lesssim -14.93$, a neighborhood of $\mathcal{F}$ becomes again linearly stable under displacements in either $\Pi^{(\tau, \alpha)}$ or $\Pi^{(\phi, \psi)}$.

Lastly, in Sect. 7, we provide some numerical illustrations of the behaviour of the normalized discriminants in $\Pi^{(\tau, \alpha)}$ and $\Pi^{(\phi, \psi)}$. We also comment on the implications of our stability theorems regarding Moeckel’s dominant mass conjecture (problem 15 in the list [2]): a relative equilibrium of the planar Newtonian N -body problem which is linearly stable must necessarily have a dominant mass. If considered for generalized potentials, it turns out that the linear stability analysis of the rhombus family imposes limitations on the potentials for which the conjecture holds, e.g., if $a \gtrsim -1.07$ or $a \lesssim -14.92$, the conjecture is false.

2 The Rhombus Family of Relative Equilibria

As in the Introduction, consider N bodies with positive masses m_i occupying positions $\mathbf{q}_i \in \mathbb{R}^d$, $i = 1, \dots, N$, and henceforth assume that $d = 2$. The bodies are subject to each other's attraction, and their motions are governed by the equations

$$m_i \ddot{\mathbf{q}}_i = \sum_{j \neq i} \mathbf{F}_{ij}, \quad i, j = 1, \dots, N, \quad (2.1)$$

where the forces $\mathbf{F}_{ij}$ are given by (1.1). The $\sum_{j \neq i} \mathbf{F}_{ij}$ are the gradients of the *potential functions* $U_a(\mathbf{q})$, which are defined as

$$U_a(\mathbf{q}) = \begin{cases} -\frac{1}{2a+2} \sum_{i < j} m_i m_j \|\mathbf{q}_i - \mathbf{q}_j\|^{2a+2}, & \text{if } a \in \mathbb{R} \setminus \{-1\}, \\ -\frac{1}{2} \sum_{i < j} m_i m_j \ln \|\mathbf{q}_i - \mathbf{q}_j\|^2, & \text{if } a = -1. \end{cases} \quad (2.2)$$

where $\mathbf{q} = (\mathbf{q}_1, \dots, \mathbf{q}_N)$, and, if $a < 0$, we assume $\mathbf{q}_i \neq \mathbf{q}_j$ for all $i \neq j$. Some values of a correspond to well-known models in Classical Mechanics: $a = 0$ is the harmonic oscillator potential, $a = -1$ is the Helmholtz-Kirchhoff Hamiltonian of the N -vortex problem, $a = -3/2$ is the Newtonian potential, and $a = -2$ is the Jacobi strong force potential. We will not discuss stability of vortices, whose dynamics obey a first order system of differential equations rather than (2.1), and will treat the case $a = -1$ just as any of the other N -body potentials in (2.2). *Relative equilibria* are solutions of the planar N -body problem (2.1) such that each body rotates with the same angular velocity ω about the center of mass $\mathbf{c} = \frac{1}{\sum_{i=1}^N m_i} \sum_{i=1}^N m_i \mathbf{q}_i$.

A simple calculation shows that in order for a configuration $\mathbf{q} = (\mathbf{q}_1, \dots, \mathbf{q}_N) \in \mathbb{R}^{2N}$ to be the initial condition of a relative equilibrium, each $\mathbf{q}_i$ must satisfy

$$\sum_{j \neq i} m_j \|\mathbf{q}_i - \mathbf{q}_j\|^{2a} (\mathbf{q}_j - \mathbf{q}_i) = \omega^2 (\mathbf{c} - \mathbf{q}_i), \quad i = 1, \dots, N. \quad (2.3)$$

We refer to (2.3) as the *relative equilibrium equations*. The form of Eq. (2.3) allows us to pick one of the masses and one of the distances to be equal to unit, as long as a suitable modification of the constant ω^2 is made in order to preserve the form of the equations.

Remark 2.1 If $a = 0$, it can be readily verified that all configurations are central and all motions are relative equilibria with angular velocity $\sqrt{\sum_{i=1}^N m_i}$. We will assume $a \neq 0$ henceforth.

In the present work we consider the *rhombus family* of relative equilibria. The elements of the rhombus family are relative equilibria of the four-body problem which interact under homogeneous forces generated by potentials (2.2) modulo homotheties and isometries of the plane. We note that, as a central configuration of four bodies of dimension two, the rhombus satisfies the equations defining a Dziobek configuration (see [1]), namely

$$\|\mathbf{q}_i - \mathbf{q}_j\|^{2a} = \xi + \eta \frac{\Delta_i}{m_i} \frac{\Delta_j}{m_j}, \quad 1 \leq i, j \leq 4, \quad i \neq j,$$

where ξ is the constant ω^2 , η is a (negative) constant, and the Δ_k are the oriented (by ordering the indices 1 through 4) areas of the triangles defined by the $\mathbf{q}_l$, $l \neq k$. A direct calculation shows that in order to satisfy the above equations, the masses at opposite vertices of the rhombus are necessarily equal.

In view of the previous paragraphs, the rhombus family consists of relative equilibria formed by points

$$\mathbf{q}_1 = (r_1, 0), \quad \mathbf{q}_2 = (0, r_2), \quad \mathbf{q}_3 = (-r_1, 0), \quad \mathbf{q}_4 = (0, -r_2),$$

where we can assume $r_1 = r, r_2 = 1$, and respective masses $m_1 = m_3 = m$ and $m_2 = m_4 = 1$. By substituting these data into the relative equilibrium equations (2.3), we find that the constant ω^2 is given by two distinct expressions

$$\omega^2 = 2[m(1+r^2)^a + 4^a] = 2[(1+r^2)^a + m4^a r^{2a}]. \quad (2.4)$$

For simplicity, let us put $s = r^2$. We solve the second equation in (2.4) for m , thus obtaining

$$m = \frac{(1+s)^a - 4^a}{(1+s)^a - 4^a s^a}. \quad (2.5)$$

Hence a rhombus relative equilibrium can be identified with a triple (a, s, m) satisfying (2.5). More formally, we define the *set of rhombus relative equilibria*

$$\mathfrak{R} = \{(a, s, m) \in \mathbb{R}^* \times \mathbb{R}_+^* \times \mathbb{R}_+^* \mid (a, s, m) \text{ satisfy Eq. (2.5)}\},$$

where $\mathbb{R}^* = \mathbb{R} - \{0\}$. The following lemmas provide some basic properties of $\mathfrak{R}$.

Lemma 2.2 *The inversion $\iota : (a, s, m) \mapsto (a, 1/s, 1/m)$ maps $\mathfrak{R}$ bijectively onto itself.*

Proof Let us make the substitution $s \mapsto 1/s$ on the right-hand side of (2.5). It follows that

$$m = \frac{(1+s)^a - 4^a}{(1+s)^a - 4^a s^a} \mapsto \frac{(1+s)^a - 4^a s^a}{(1+s)^a - 4^a} = \frac{1}{m}.$$

□

Lemma 2.2 has an important consequence for the description of rhombus relative equilibria. The orientation preserving Möbius transformation

$$s = \frac{2+v}{2-v} \quad (2.6)$$

maps the interval $[-2, 2]$ onto the extended nonnegative semi-axis $[0, \infty]$, and converts the inversion $s \mapsto 1/s$ into a reflection $v \mapsto -v$. Under this transformation, Eq. (2.5) assumes the simpler form

$$m = \frac{1 - (2-v)^a}{1 - (2+v)^a}, \quad -2 < v < 2, \quad a \in \mathbb{R}^*. \quad (2.7)$$

The above equation will prove very useful throughout the article.

Remark 2.3 We can also perform the corresponding Möbius transformation on m , and in this way express a new mass parameter as an odd function of v . This will be done in order to produce Fig. 2, however, since transforming m produces an expression not as simple as (2.7), such transformation will not be used elsewhere in the paper.

Lemma 2.4 *A rhombus relative equilibrium must be such that $1/3 < s < 3$.*

Proof The Möbius transformation (2.6) maps the open interval $(-1, 1)$ onto $(1/3, 3)$. Consider Eq. (2.7). If $1 < v < 2$, the numerator and denominator of m have opposite signs. Using the reflection symmetry $v \mapsto -v$, we conclude that m positive imposes $v \in (-1, 1)$.

□

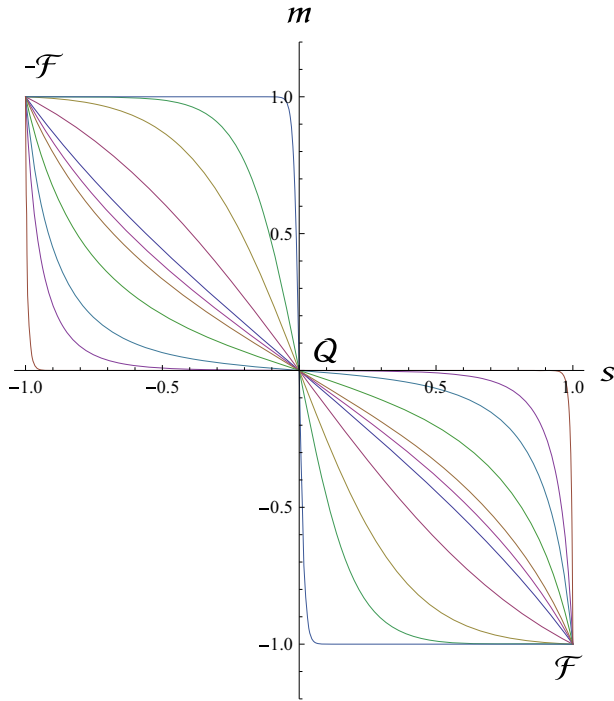


Fig. 2 Some graphs in the pencil of rhombus relative equilibria, after suitable Möbius transformations. Notice the three focal points Q , F and $F' = -F$

Remark 2.5 The bounds $1/3$ and 3 for the ratio of the squares of the diagonals in Lemma 2.4 appear in [6]. These bounds actually hold for the squares of the mutual distances in arbitrary convex four-body relative equilibria [7, 11].

Remark 2.6 It is sometimes convenient to consider masses which are zero or infinity. We will do so accordingly.

Lemma 2.7 For every fixed a , Eq. (2.5) defines a decreasing function $m = m_a(s)$ of $s \in (1/3, 3)$.

Proof Recall (2.6) is an orientation preserving transformation. Fix a and define $m = m_a(v)$ by (2.7) with $v \in (-1, 1)$. The numerator of the derivative of $m_a(v)$ is the expression

$$a \{ (2-v)^{a-1} [1 - (2+v)^a] + (2+v)^{a-1} [1 - (2-v)^a] \},$$

which is negative for all $a \in \mathbb{R}^*$. $\square$

Lemma 2.8 For every fixed $1 < s < 3$, Eq. (2.5) defines a decreasing function $m = m_s(a)$ on $\mathbb{R}^*$.

Proof Fixing $1 \leq s < 3$ corresponds to fixing $0 \leq v < 1$. Let $f(a) = 1 - (2-v)^a$ and $g(a) = 1 - (2+v)^a$. We have that $m_s(a) = f(a)/g(a)$, so the sign of the derivative of $m_s(a)$ is the same as the sign of the difference $f'(a)g(a) - f(a)g'(a)$, which is given by

$$-(2-v)^a [1 - (2+v)^a] \ln(2-v) + [1 - (2-v)^a] (2+v)^a \ln(2+v).$$

It follows that every a is a critical point if $v = 0$. Excluding the case $v = 0$, a critical point of $m_s(a)$ must be a solution of the equation

$$\frac{(2-v)^a \ln(2-v)}{(2-v)^a - 1} = \frac{(2+v)^a \ln(2+v)}{(2+v)^a - 1}.$$

For $0 < v < 1$ we have $2-v \neq 2+v$, so the equation above holds if and only if

$$h(t) = \frac{t^a \ln t}{t^a - 1}, \quad \text{for } 1 < t < 3, t \neq 2,$$

is not an injective function. We will show this is false, so $m_s(a)$ cannot have any critical points. We have that

$$h'(t) = \frac{t^{a-1}}{(t^a - 1)^2} l(t), \quad \text{with } l(t) = -1 + t^a - a \ln t.$$

Consider $l(t)$ defined on $1 \leq t < 3$; from $l'(t) = -\frac{a}{t}(1 - t^a)$, we deduce that $l(1) = 0$ is the minimum of $l(t)$. Thus $l(t)$ is positive if $1 < t < 3$, $t \neq 2$, and $h(t)$ is a strictly increasing function, so $m_s(a)$ cannot have a critical point.

In order to finish the proof, we determine the sign of $m'_s(a)$ at $a = -1$ and $a = 1$. We compute

$$f'(-1)g(-1) - f(-1)g'(-1) = \frac{(1+v) \ln(2-v) + (-1+v) \ln(2+v)}{v^2 - 4}.$$

The numerator of the right-hand side expression vanishes at $v = 0, 1$ and has second derivative $\frac{2v(v^2-16)}{(v^2-4)^2}$, which is strictly negative on $0 \leq v \leq 1$. It follows that $m'_s(-1) < 0$ for $1 < s < 3$ and $a < 0$. Now consider

$$f'(1)g(1) - f(1)g'(1) = (2+v-v^2) \ln(2-v) + (-2+v+v^2) \ln(2+v).$$

The expression on the right-hand side vanishes at $v = 0, 1$, has second derivative $\frac{6v}{v^2-4} + 4 \tanh^{-1}\left(\frac{v}{2}\right)$, and third derivative $-2\frac{v^2-28}{(v^2-4)^2}$, so it is negative on $0 < v < 1$. Thus $m'_s(a)$ is negative for $1 < s < 3$ and $a > 0$. $\square$

Lemmas 2.2, 2.4, 2.7 and 2.8 determine the structure of the family of rhombus relative equilibria.

Theorem 2.9 *The family of graphs defined by (2.5) for $a \in \mathbb{R}^*$, and $m = m_0(s) = \ln\left(\frac{1+s}{4}\right) / \ln\left(\frac{1+s}{4s}\right)$ if $a = 0$, defines a pencil of curves parametrized by the exponent a which foliates the interior of the rectangles $[1/3, 1] \times [1, \infty]$ and $[1, 3] \times [0, 1]$ in the extended (s, m) -plane. The pencil of rhombus relative equilibria graphs has precisely three focal points, namely $\mathcal{Q} = (1, 1)$, which corresponds to the square with equal masses at its vertices, the oblate rhombus $\mathcal{F} = (3, 0)$ with a pair of opposite zero masses, and the prolate rhombus $\mathcal{F}' = (1/3, \infty)$ with a pair of opposite infinite masses. Let $\tilde{\mathfrak{R}}$ be the projection of the set of rhombus relative equilibria $\mathfrak{R}$ onto the extended (s, m) -plane $\tilde{\mathbb{R}}^2 = [-\infty, \infty] \times [-\infty, \infty]$. The boundary of $\tilde{\mathfrak{R}}$ in $\tilde{\mathbb{R}}^2$ is the union of the graph of $m_0(s)$ and the polygonals*

$$\{(1/3, m) / m \geq 1\} \cup \{(s, 1) / 1/3 \leq s \leq 3\} \cup \{(3, m) / 0 \leq m \leq 1\},$$

and

$$\{(s, \infty) / 1/3 \leq s \leq 1\} \cup \{(1, m) / m \geq 0\} \cup \{(s, 0) / 1 \leq s \leq 3\}.$$

which correspond to the limiting cases $a = -\infty$ and $a = \infty$, respectively.

Proof The claims follow directly from Eq. (2.5) and the lemmas. The focal points were determined by substituting the values $s = 1/3, 1, 3$ into (2.5), and the non-existence of additional focal points follows from Lemma 2.8. $\square$

Figure 2 depicts graphs in $\tilde{\mathfrak{R}}$ for $a = \pm\frac{1}{2}, \pm 2, \pm 5, \pm 10, \pm 100$ after applying the transformations $m \mapsto (m - 1)/(m + 1)$ and $s \mapsto 2(s - 1)/(s + 1)$.

3 Linear Stability

This section is mostly a preparation for the linear stability analysis of the rhombus relative equilibria, although we also provide some general results on normal linear stability. In the introductory paragraphs and first subsection we follow closely the presentation in [8].

Consider $\mathbb{R}^3$ with the usual Cartesian coordinates (x, y, z) and assume the (x, y) -plane contains the relative equilibrium. By changing to coordinates with respect to a reference frame in uniform rotation around the z -axis with angular velocity ω , and denoting the corresponding new coordinates of $\mathbb{R}^{3N}$ by $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_N)$, the equations of motion (2.1) assume the form

$$\hat{M}\ddot{\mathbf{x}} + 2\omega\hat{J}\hat{M}\dot{\mathbf{x}} = \nabla U_a(\mathbf{x}) + \omega^2\hat{M}\hat{I}\mathbf{x}, \quad (3.1)$$

where $\hat{M}$, $\hat{I}$ and $\hat{J}$ are the block-diagonal matrices with blocks

$$\begin{bmatrix} m_k & 0 & 0 \\ 0 & m_k & 0 \\ 0 & 0 & m_k \end{bmatrix}, \quad k = 1, \dots, N, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

along the respective main diagonals.

A relative equilibrium of the N -body problem is a constant solution $\mathbf{x}^{(0)}$ of Eq. (3.1) for some fixed ω^2 . We say that $\mathbf{x}^{(0)}$ is *linearly stable* if the origin is a stable solution of the linearization of Eq. (3.1) at $\mathbf{x}^{(0)}$:

$$\hat{M}\ddot{\delta} + 2\omega\hat{J}\hat{M}\dot{\delta} = (D\nabla U_a(\mathbf{x}^{(0)})) + \omega^2\hat{M}\hat{I}\delta, \quad (3.2)$$

where $\delta = (\delta_1, \dots, \delta_N) \in \mathbb{R}^{3N}$ represents a displacement of the origin. Actually, due to the invariance of the potential functions $U_a(\mathbf{x})$ under rigid motions and the behaviour of $U_a(\mathbf{x})$ under homotheties, the origin is never linearly stable, so in order to discuss linear stability of relative equilibria, we firstly mod out the subspace of the space of displacements which infinitesimally represent such transformations.

Let us view (3.2) as a system of N linear differential equations in the δ_j . By collecting all the z components of the individual displacements δ_j , Eq. (3.2) decouple into two sets of equations, one for the planar displacements, and the other for the normal displacements. Indeed, the decoupling follows from the block-structure of $D\nabla U_a(\mathbf{x})$, which consists of 3×3 blocks of the form

$$\begin{bmatrix} B_{jk} & O_{2 \times 1} \\ O_{1 \times 2} & C_{jk} \end{bmatrix},$$

where the O s represent null matrices. In the above matrix, the blocks B_{jk} are of size 2×2 and are given by

$$m_j m_k r_{jk}^{2a} \left(2a u_{jk} u_{jk}^T + I \right), \quad \text{if } j \neq k, \quad \text{and} \quad - \sum_{l \neq k} B_{kl}, \quad \text{if } j = k, \quad (3.3)$$

where $u_{jk} = \frac{\mathbf{x}_k - \mathbf{x}_j}{r_{jk}}$, and $r_{jk} = \|\mathbf{x}_j - \mathbf{x}_k\|$. The entries C_{jk} are

$$C_{jk} = m_j m_k r_{jk}^{2a}, \quad \text{if } j \neq k, \quad \text{and} \quad - \sum_{l \neq k} C_{kl}, \quad \text{if } j = k. \quad (3.4)$$

3.1 Normal Linear Stability

Let C be the $N \times N$ (symmetric) matrix whose entries are C_{jk} , and let $\delta_z \in \mathbb{R}^N$ denote the vector formed by the z coordinates of each displacement δ_j , $j = 1, \dots, N$. The differential equations for the normal displacements from the relative equilibrium $\mathbf{x}^{(0)}$ are:

$$M_z \ddot{\delta}_z = C \delta_z, \quad (3.5)$$

where M_z is the diagonal matrix $\text{diag}(m_1, \dots, m_N)$. It is clear that the origin is a stable solution of Eq. (3.5) if and only if $M_z^{-1}C$ is a diagonalizable matrix and its eigenvalues are all real and negative. However, the vector $(1, \dots, 1)$, which corresponds to vertical translations of the relative equilibrium, is always in the kernel of C . We say that the relative equilibrium $\mathbf{x}^{(0)}$ is *normally nondegenerate* if $(1, \dots, 1)$ generates the kernel of C . We say that $\mathbf{x}^{(0)}$ is *normally linearly stable* if $\mathbf{x}^{(0)}$ is normally nondegenerate, $M_z^{-1}C$ is diagonalizable and the nonzero eigenvalues of $M_z^{-1}C$ are all real and negative.

Proposition 3.1 *Every relative equilibrium of the N -body problem with potentials (2.2) is normally linearly stable.*

Proof The matrix $\tilde{C} = M_z^{-1}C$ is symmetric with respect to the inner product defined by M_z . Indeed, we have that

$$\tilde{C}^T M_z = (M_z^{-1}C)^T M_z = C^T = C = M_z(M_z^{-1}C) = M_z \tilde{C}.$$

Thus the matrix $\tilde{C}$ is diagonalizable (by an M_z -orthogonal basis) and that all its eigenvalues are real.

Recall the expressions (3.4) for C_{jk} and that $C_{kk} = -\sum_{l \neq k} C_{kl}$. Let λ be an eigenvalue of $\tilde{C}$ and let $v \neq 0$ an eigenvector of $\tilde{C}$ associated with λ . Thus $Cv = \lambda Mv$. Let v_k be the coordinate of v with maximum absolute value. By replacing v with $-v$ if necessary, we may suppose that $v_k > 0$. It follows that

$$\lambda m_k v_k = \sum_{j=1}^n C_{kj} v_j \leq C_{kk} v_k + \sum_{j \neq k} |C_{kj} v_j| \leq (C_{kk} + \sum_{j \neq k} |C_{kj}|) v_k = 0.$$

Since $m_k > 0$, we conclude that $\lambda \leq 0$.

Now let v be a non-zero vector in the kernel of $\tilde{C}$. Thus $Cv = 0$, and we have that

$$C_{kk} v_k + \sum_{j \neq k} C_{kj} v_j = \sum_{j \neq k} C_{kj} (v_j - v_k) = 0, \quad k = 1, \dots, N.$$

Let v_k be the coordinate of v with maximum value. Since $C_{kj} > 0$, for all $j \neq k$, we have that $v_j - v_k \leq 0$, thus $v_j - v_k = 0$, for all j, k . Therefore v is a multiple of $(1, \dots, 1)$. $\square$

3.2 Planar Linear Stability of the Rhombus

In order to study the planar linear stability of a relative equilibrium $\mathbf{x}^{(0)}$, we must analyze the roots of the *planar stability polynomial*

$$P(\lambda) = \det(M^{-1}B + (\omega^2 - \lambda^2)I - 2\omega\lambda J), \quad (3.6)$$

which is deduced from Eq. (3.2). On the right-hand side of the above equation, B is the matrix corresponding to the restriction of $D\nabla U_a(\mathbf{x}^{(0)})$ to the subspace of planar displacements, I denotes the identity matrix, and M and J denote the $2N \times 2N$ block-diagonal matrices with blocks

$$\begin{bmatrix} m_k & 0 \\ 0 & m_k \end{bmatrix}, \quad k = 1, \dots, N, \quad \text{and} \quad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

along the respective main diagonals. In the following paragraphs, we use the results of “Appendix A” to produce a full factorization of $P(\lambda)$ when the relative equilibrium is a rhombus.

We compute the matrix B at the rhombus relative equilibrium

$$\mathbf{x}^{(0)} = (r, 0, 0, 1, -r, 0, 0, -1),$$

with masses $m_1 = m_3 = m$ and $m_2 = m_4 = 1$. We noted in the introductory discussion that B has a (2×2) -block structure whose blocks B_{jk} are given by (3.3). At $\mathbf{x}^{(0)}$, we have that

$$B = \begin{bmatrix} B_{11} & B_{12} & B_{13} & B_{14} \\ B_{12} & B_{22} & B_{14} & B_{24} \\ B_{13} & B_{14} & B_{11} & B_{12} \\ B_{14} & B_{24} & B_{12} & B_{22} \end{bmatrix},$$

where

$$B_{12} = m(1 + r^2)^{a-1} \begin{bmatrix} (2a + 1)r^2 + 1 & -2ar \\ -2ar & 1 + 2a + r^2 \end{bmatrix},$$

$$B_{14} = m(1 + r^2)^{a-1} \begin{bmatrix} (2a + 1)r^2 + 1 & 2ar \\ 2ar & 1 + 2a + r^2 \end{bmatrix},$$

and

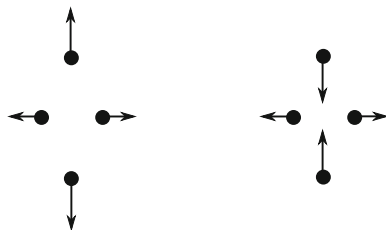
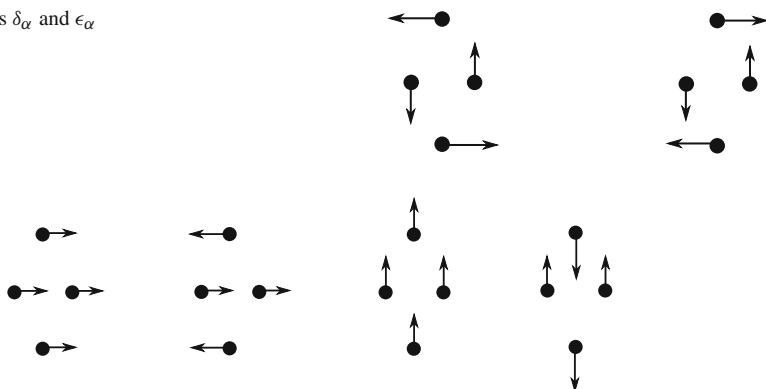
$$B_{13} = m^2(2r)^{2a} \begin{bmatrix} 2a + 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad B_{24} = 2^{2a} \begin{bmatrix} 1 & 0 \\ 0 & 2a + 1 \end{bmatrix}.$$

The diagonal blocks are simply

$$B_{11} = -B_{12} - B_{13} - B_{14}, \quad B_{22} = -B_{12} - B_{14} - B_{24}.$$

Next we apply $M^{-1}B$ to each of the vectors (A.1) through (A.8) constructed in “Appendix A” and illustrated in Figs. 3, 4 and 5. The results are as follows

$$\begin{aligned} M^{-1}B\delta_\tau &= -(2a + 1)\omega^2\delta_\tau, & M^{-1}B\delta_\alpha &= -\omega^2\delta_\alpha, \\ M^{-1}B\delta_\phi &= 0, & M^{-1}B\delta_\psi &= 0, \\ M^{-1}B\epsilon_\tau &= \lambda_1^{(\tau, \alpha)}\epsilon_\tau, & M^{-1}B\epsilon_\alpha &= \lambda_2^{(\tau, \alpha)}\epsilon_\alpha, \\ M^{-1}B\epsilon_\phi &= \lambda_1^{(\phi, \psi)}\epsilon_\phi, & M^{-1}B\epsilon_\psi &= \lambda_2^{(\phi, \psi)}\epsilon_\psi, \end{aligned}$$

Fig. 3 The vectors δ_τ and ϵ_τ

Fig. 4 The vectors δ_α and ϵ_α

Fig. 5 The vectors δ_ϕ , ϵ_ϕ and δ_ψ , ϵ_ψ

where in terms of the parameter $s = r^2$, we have

$$\lambda_1^{(\tau, \alpha)} = -4a [m4^a s^a - (m-1)s(1+s)^{a-1}] - \omega^2 \quad (3.7)$$

$$\lambda_2^{(\tau, \alpha)} = -4a(1+ms)(1+s)^{a-1} - \omega^2, \quad (3.8)$$

$$\lambda_1^{(\phi, \psi)} = -2(m+1)(1+s)^{a-1} [1 + (2a+1)s], \quad (3.9)$$

$$\lambda_2^{(\phi, \psi)} = -2(m+1)(1+s)^{a-1}(1+2a+s). \quad (3.10)$$

The eigenvalues $-(2a+1)\omega^2$, $-\omega^2$, 0 , 0 arise from the classical symmetries of the generalized N -body problem and the homogeneity of the corresponding potential functions. These eigenvalues are shared by all relative equilibria, and, together with the associated eigenvectors, are considered in the stability analysis. By construction, the remaining four eigenvectors of $M^{-1}B$ satisfy the relations $\epsilon_\tau = -J\epsilon_\alpha$, $\epsilon_\phi = -J\epsilon_\psi$, where J is the 8×8 block-diagonal matrix with blocks $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$ along its main diagonal. Thus, the planes $\Pi^{(\tau, \alpha)}$ and $\Pi^{(\phi, \psi)}$ formed by linear combinations of ϵ_τ , ϵ_α and ϵ_ϕ , ϵ_ψ , respectively, are simultaneously invariant by the matrices $M^{-1}B$ and J , and the factors of the stability polynomial corresponding to $\Pi^{(\tau, \alpha)}$ and $\Pi^{(\phi, \psi)}$ determine the linear stability of the rhombus relative equilibria.

Let λ_1, λ_2 denote the respective eigenvalues of the pairs $\epsilon_\tau, \epsilon_\alpha$ or $\epsilon_\phi, \epsilon_\psi$. The planar stability polynomial (3.6) has two quadratic factors in $\varkappa = \omega^2 - \lambda^2$ of the form

$$\begin{vmatrix} \lambda_1 + \varkappa & 2\omega\lambda \\ -2\omega\lambda & \lambda_2 + \varkappa \end{vmatrix} = \varkappa^2 + (\lambda_1 + \lambda_2 - 4\omega^2)\varkappa + \lambda_1\lambda_2 + 4\omega^4.$$

In terms of λ^2 , the roots of the above polynomial are given by

$$\lambda^2 = \frac{(\lambda_1 + \lambda_2 - 2\omega^2) \pm \sqrt{(\lambda_1 + \lambda_2 - 4\omega^2)^2 - 4(\lambda_1\lambda_2 + 4\omega^4)}}{2}. \quad (3.11)$$

Note the right-hand side of (3.11) can be expressed in terms of the sum and the difference of λ_1 and λ_2 , for

$$(\lambda_1 + \lambda_2 - 4\omega^2)^2 - 4(\lambda_1\lambda_2 + 4\omega^4) = (\lambda_1 - \lambda_2)^2 - 8\omega^2(\lambda_1 + \lambda_2).$$

Let

$$\Delta = (\lambda_1 - \lambda_2)^2 - 8\omega^2(\lambda_1 + \lambda_2). \quad (3.12)$$

We list three conditions which together are necessary and sufficient for λ^2 to be real and negative:

$$\lambda_1 + \lambda_2 - 2\omega^2 < 0, \quad (\lambda_1 + \lambda_2 - 2\omega^2)^2 - \Delta > 0, \quad \Delta \geq 0. \quad (3.13)$$

The second condition in (3.13) has a nice alternative expression, since

$$(\lambda_1 + \lambda_2 - 2\omega^2)^2 - \Delta = (\lambda_1 + \lambda_2 + 2\omega^2)^2 - (\lambda_1 - \lambda_2)^2 = 4(\lambda_1 + \omega^2)(\lambda_2 + \omega^2).$$

We extract from inequalities (3.13) some sets of simpler conditions.

Proposition 3.2 *The following are three sets of sufficient conditions for λ^2 to be real and negative:*

- (I) $\lambda_1 + \omega^2 < 0$ and $\lambda_2 + \omega^2 < 0$.
- (II) $\lambda_1 + \omega^2 > 0$, $\lambda_2 + \omega^2 > 0$ and $\Delta > 0$.
- (III) $\lambda_1 + \lambda_2 - 2\omega^2 < 0$ and $\Delta = 0$.

If $\Delta < 0$ or if $\lambda_1 + \omega^2$ and $\lambda_2 + \omega^2$ have opposite signs, then λ^2 is nonreal.

Proof Let us make the change of variables $\mu_1 = \lambda_1 + \lambda_2$, $\mu_2 = \lambda_1 - \lambda_2$. In terms of the variables μ_1 , μ_2 , inequalities (3.13) are

$$\mu_1 - 2\omega^2 < 0, \quad (\mu_1 + \mu_2 + 2\omega^2)(\mu_1 - \mu_2 + 2\omega^2) > 0, \quad \mu_2^2 - 8\omega^2\mu_1 \geq 0.$$

The second inequality can be rewritten as two pairs of inequalities

$$\mu_1 \pm \mu_2 + 2\omega^2 > 0, \quad \text{and} \quad \mu_1 \pm \mu_2 + 2\omega^2 < 0.$$

If represented in the (μ_1, μ_2) -plane, the former pair of inequalities determines an open quadrant which we denote by Q_+ , while the latter pair defines an open quadrant denoted by Q_- .

We observe that the parabola $\mu_2^2 = 8\omega^2\mu_1$ is tangent to the boundaries of quadrant Q_+ at the points $(2\omega^2, \pm 4\omega^2)$, and that its remaining points lie within Q_+ . Indeed, at the parabola we have

$$\mu_1 \pm \mu_2 + 2\omega^2 = \frac{1}{8\omega^2}(\mu_2 \pm 4\omega^2)^2 \geq 0.$$

In particular, we also have that $\mu_1 \pm \mu_2 + 2\omega^2 < 0$ lies within the region determined by $\mu_2^2 - 8\omega^2\mu_1 \geq 0$ and $\mu_1 - 2\omega^2 < 0$. This proves condition (I), which is equivalent to having $(\mu_1, \mu_2) \in Q_-$. Condition (II) corresponds to the bounded region of Q_+ limited by the parabola $\mu_2^2 = 8\omega^2\mu_1$. Notice that, in either case, condition $\mu_1 - 2\omega^2 < 0$ can be dropped from (3.13). Condition (III) corresponds to the arc of the parabola $\mu_2^2 = 8\omega^2\mu_1$ contained in Q_+ and such that $0 \leq \mu_1 < 2\omega^2$. The subsequent claims are immediately verified. $\square$

Proposition 3.3 Consider displacements in either of the subspaces

$$\Pi^{(\tau, \alpha)} = \text{Span}\{\epsilon_\tau, \epsilon_\alpha\} \quad \text{or} \quad \Pi^{(\phi, \psi)} = \text{Span}\{\epsilon_\phi, \epsilon_\psi\}.$$

A rhombus relative equilibrium is linearly stable if and only if either conditions (I) or conditions (II) in Proposition 3.2 are fulfilled. Conditions (III) imply the rhombus relative equilibrium is unstable (albeit it is still spectrally stable).

Proof The following argument was inspired by remarks in [9]. If $\Delta > 0$ for either plane $\Pi^{(\tau, \alpha)}$ or $\Pi^{(\phi, \psi)}$, the planar stability matrix has distinct eigenvalues, so a basis of eigenvectors can be produced and the relative equilibrium is linearly stable as long as conditions (I) or conditions (II) in Proposition 3.2 hold. If $\Delta = 0$, the planar stability matrix has repeated eigenvalues. However, the 2×2 blocks corresponding to $\Pi^{(\tau, \alpha)}$ or $\Pi^{(\phi, \psi)}$ cannot have nullity two if conditions (III) hold because, from Eq. (3.11), the inequality $\lambda_1 + \lambda_2 - 2\omega^2 < 0$ implies $\lambda \neq 0$. $\square$

Remark 3.4 A rhombus relative equilibrium is never degenerate in the sense of [8], i.e., the $(\lambda_1 + \omega^2)(\lambda_2 + \omega^2)$ never vanishes, as we will see in Sects. 5 and 6.

4 Linear Stability of the Square

According to Proposition 2.9, the square $\mathcal{Q} = (1, 1)$ is the only focal point with positive finite mass ratio of the pencil $\mathfrak{R}$ formed by the curves of rhombus relative equilibria. In this brief section we analyze the linear stability of $\mathcal{Q}$.

Recall Propositions 3.2 and 3.3. We have that

$$\begin{aligned} \lambda_1^{(\tau, \alpha)} + \omega^2 &= -4^{1+a}a, \quad \lambda_2^{(\tau, \alpha)} + \omega^2 = -2^{2+a}a, \\ \lambda_1^{(\phi, \psi)} + \omega^2 &= \lambda_2^{(\phi, \psi)} + \omega^2 = 2^{1+a}(2^a - 2a - 1) \end{aligned}$$

and the discriminants in $\Pi^{(\tau, \alpha)}$ and $\Pi^{(\phi, \psi)}$ are, respectively,

$$\begin{aligned} \Delta_{\diamond}^{(\tau, \alpha)} &= 4^{2+a}[(2^a - 1)^2a^2 + 4(2^a + 1)^2(1 + a)], \\ \Delta_{\diamond}^{(\phi, \psi)} &= 2^{7+2a}(1 + a)(1 + 2^a). \end{aligned}$$

$\Delta_{\diamond}^{(\tau, \alpha)}$ possesses an interesting factorization, namely

$$\Delta_{\diamond}^{(\tau, \alpha)} = 4^{2+a}(2^a - 1)^2 f_1(a) f_2(a),$$

where

$$f_1(a) = a + 2 \frac{(2^a + 1)}{(1 + 2^{a/2})^2}, \quad \text{and} \quad f_2(a) = a + 2 \frac{(2^a + 1)}{(-1 + 2^{a/2})^2}.$$

We compute the derivatives

$$f_1'(a) = 1 + \frac{2^{1+a/2}(-1 + 2^{a/2}) \ln 2}{(1 + 2^{a/2})^3}, \quad f_2'(a) = 1 - \frac{2^{1+a/2}(1 + 2^{a/2}) \ln 2}{(-1 + 2^{a/2})^3}.$$

If $a < 0$, we have $1 > 2^{a/2}$, so $f_2'(a) > 0$. We also have $f_1'(a) > 0$, for if we write $f_1'(a)$ as a rational function of $b = 2^{a/2}$, the denominator is positive and the numerator is also positive because

$$(b + 1)^3 + 2b(b - 1) \ln 2 = b^3 + (3 + 2 \ln 2)b^2 + (3 - 2 \ln 2)b + 1 > 0.$$

Thus $f_1(a)$ and $f_2(a)$ have at most one zero each on $a < 0$. Direct calculations give the following results:

$$f_1(-1) = 17 - 12\sqrt{2} > 0, \quad f_1(-6/5) = 4 \frac{4 - 6 \cdot 2^{2/5} + 2^{4/5}}{5(2 + 2^{2/5})^2} < 0.$$

and

$$f_2(-19/5) = \frac{-144 + 152 \cdot 2^{1/10} - 9 \cdot 2^{1/5}}{5(4 + 2^{1/10})^2} > 0, \quad f_2(-4) = -\frac{2}{9}.$$

Let $-6/5 < a_{\diamond}^{(1)} < -1$ and $-4 < a_{\diamond}^{(2)} < -19/5$ be the respective zeros of $f_1(a)$ and $f_2(a)$. Notice $\Delta_{\diamond}^{(\tau, \alpha)} \leq 0$ if $a_{\diamond}^{(2)} \leq a \leq a_{\diamond}^{(1)}$.

Proposition 4.1 *The square relative equilibrium $\mathcal{Q}$ is linearly stable if $a > -1$. For $a \leq -1$, there exist numbers $-6/5 < a_{\diamond}^{(1)} < -1$ and $-4 < a_{\diamond}^{(2)} < -19/5$ such that, under displacements in $\Pi^{(\tau, \alpha)}$, $\mathcal{Q}$ is linearly stable if $a > a_{\diamond}^{(1)}$ or $a < a_{\diamond}^{(2)}$, and is unstable for the remaining values of a . Under displacements in $\Pi^{(\phi, \psi)}$, $\mathcal{Q}$ is unstable if $a \leq -1$.*

Proof The claims follow from Propositions 3.2 and 3.3 and the analysis of $\Delta_{\diamond}^{(\tau, \alpha)}$ above. $\square$

In the next two sections we analyze the linear stability of arbitrary rhombus relative equilibria.

5 Linear Stability under Displacements in $\Pi^{(\tau, \alpha)}$

Recall the expressions (3.7), (3.8) and (3.12). Corresponding to $\Pi^{(\tau, \alpha)}$, we have

$$\begin{aligned} \lambda_1^{(\tau, \alpha)} + \omega^2 &= -4a [m4^a s^a - (m-1)s(1+s)^{a-1}], \\ \lambda_2^{(\tau, \alpha)} + \omega^2 &= -4a(1+ms)(1+s)^{a-1}, \end{aligned}$$

and

$$\Delta^{(\tau, \alpha)} = 16 \left\{ \omega^4(1+a) + [(s-1)(1+s)^{a-1} + m(4^a s^a - 2s(1+s)^{a-1})]^2 a^2 \right\}.$$

Lemma 5.1 *At any rhombus relative equilibrium we have*

$$\lambda_1^{(\tau, \alpha)} + \omega^2 > 0, \quad \text{and} \quad \lambda_2^{(\tau, \alpha)} + \omega^2 > 0, \quad \text{if } a < 0,$$

and

$$\lambda_1^{(\tau, \alpha)} + \omega^2 < 0, \quad \text{and} \quad \lambda_2^{(\tau, \alpha)} + \omega^2 < 0, \quad \text{if } a > 0.$$

Proof The inequalities for $\lambda_2^{(\tau, \alpha)} + \omega^2$ are clear, and at $\mathcal{F} = (3, 0)$ we have that $\lambda_1^{(\tau, \alpha)} + \omega^2 = -3a4^a$. We apply the Möbius transformation (2.6) and express $\lambda_1^{(\tau, \alpha)} + \omega^2$ as a function of v . After simplifications, we obtain

$$\lambda_1^{(\tau, \alpha)} + \omega^2 = -a4^a \frac{(2+v)^a}{1-(2+v)^a} [(2+v)^{1-a} + (2-v)^{1-a} - 4], \quad -1 < v < 1.$$

The expression multiplying the square brackets is positive for every $a \in \mathbb{R}^*$. The expression in the square brackets defines an even function of v , has value $4(2^{-a} - 1)$ at $v = 0$, and has derivative

$$(1 - a) [(2 + v)^{-a} - (2 - v)^{-a}]. \quad (5.1)$$

The above expression is positive if $a < 0$ and $0 < v < 1$, negative if $0 < a < 1$ and $0 < v < 1$, and vanishes at $a = 1$. If $a > 1$, (5.1) is positive, but at $v = 1$, $(2 + v)^{1-a} + (2 - v)^{1-a} - 4$ is negative. The lemma is proved. $\square$

Proposition 5.2 *If $a > -1$, every rhombus relative equilibrium is linearly stable with respect to displacements in the subspace $\Pi^{(\tau, \alpha)}$.*

Proof The statement is a direct consequence of Proposition 3.3 and the above lemma. $\square$

In view of Propositions 3.2, 3.3 and Lemma 5.1, if $a < 0$ then linear stability can only occur if the discriminant $\Delta^{(\tau, \alpha)}$ is positive. We proceed to a detailed analysis of $\Delta^{(\tau, \alpha)}$.

5.1 Analysis of $\Delta^{(\tau, \alpha)}$

Throughout this subsection we suppose $a < 0$. From the second expression in (2.4) for ω^2 , we deduce

$$(s - 1)(1 + s)^{a-1} + m[4^a s^a - 2s(1 + s)^{a-1}] = \frac{\omega^2}{2} - v^2,$$

where $v^2 = 2(1 + ms)(1 + s)^{a-1}$. Hence we can write

$$\Delta^{(\tau, \alpha)} = 16[\omega^4(1 + a) + X^2 a^2], \quad \text{with } X^2 = \frac{1}{4}(\omega^2 - 2v^2)^2. \quad (5.2)$$

Our strategy is to understand how v^2 and ω^2 relate to each other.

Lemma 5.3 *The ratio v^2/ω^2 is invariant under the inversion $\iota : (a, s, m) \mapsto (a, 1/s, 1/m)$. Hence the signs of $\Delta^{(\tau, \alpha)}$ at (a, s, m) and $(a, 1/s, 1/m)$ are the same.*

Proof Let $\tilde{v} = v \circ \iota$ and $\tilde{\omega} = \omega \circ \iota$. Simple calculations show that

$$\tilde{v}^2 = m^{-1} s^{-a} v^2, \quad \text{and} \quad \tilde{\omega}^2 = m^{-1} s^{-a} \omega^2,$$

where we have made use of the expressions (2.4) of ω^2 . The lemma follows. $\square$

Lemma 5.4 $\omega^2 > v^2$.

Proof A direct computation produces

$$\omega^2 - v^2 = 2[m4^a s^a - (m - 1)s(1 + s)^{a-1}] = \frac{\lambda_1^{(\tau, \alpha)} + \omega^2}{-4a}$$

and the latter is positive according to Lemma 5.1. $\square$

Corollary 5.5 $\omega^4 > 4X^2$.

Proof We verify that $\omega^4 - 4X^2 = 4v^2(\omega^2 - v^2)$, or, equivalently,

$$\frac{X^2}{\omega^4} = \frac{1}{4} - \frac{v^2}{\omega^2} \left(1 - \frac{v^2}{\omega^2}\right), \quad (5.3)$$

and apply Lemma 5.4. $\square$

It is possible to sharpen the estimate in the above corollary. For that we determine an upper bound for the ratio X^2/ω^4 at a rhombus relative equilibrium (a, s, m) with respect only to the variable s . From Eq. (5.3), using the abbreviations $\max$ and $\min$ to indicate maximum and minimum with respect to s , we have that

$$\max \left\{ \frac{X^2}{\omega^4} \right\} = \frac{1}{4} - \min \left\{ \frac{v^2}{\omega^2} \left(1 - \frac{v^2}{\omega^2} \right) \right\}. \quad (5.4)$$

Proposition 5.6 $\max \left\{ \frac{X^2}{\omega^4} \right\} = \begin{cases} 1/16, & \text{if } -\ln 3/\ln 2 \leq a < 0, \\ (2^a - 1)^2/[4(2^a + 1)^2], & \text{if } a < -\ln 3/\ln 2. \end{cases}$

Proof Let

$$Y^2 = \omega^2 - v^2 = 2[m4^a s^a - (m-1)s(1+s)^{a-1}].$$

It is clear that $\frac{v^2}{\omega^2} \left(1 - \frac{v^2}{\omega^2} \right) = \frac{Y^2}{\omega^2} \left(1 - \frac{Y^2}{\omega^2} \right)$. If we replace s and m with the expressions (2.6) and (2.7), we obtain, after simple manipulations,

$$\frac{Y^2}{\omega^2} = \frac{(2+v)^{1-a} + (2-v)^{1-a} - 4}{4[(4-v^2)^{-a} - 1]},$$

where in view of Lemma 5.3, we may restrict the range of v from $-1 < v < 1$ to $0 \leq v < 1$. Since $a < 0$, the numerator and the denominator of the expression on the right-hand side are positive, and the numerator is an increasing function of v , while the denominator is a decreasing function of v . We conclude that Y^2/ω^2 increases monotonically between the values

$$\min \left(\frac{Y^2}{\omega^2} \right) = \frac{2^a}{1+2^a}, \quad \max \left(\frac{Y^2}{\omega^2} \right) = \frac{3}{4}.$$

Thus we have that

$$\min \left[\frac{Y^2}{\omega^2} \left(1 - \frac{Y^2}{\omega^2} \right) \right] = \min \left\{ \frac{2^a}{(1+2^a)^2}, \frac{3}{16} \right\},$$

and the right-hand side is $2^a/[(1+2^a)^2]$ if $a < -\ln 3/\ln 2$, and is $3/16$ if $-\ln 3/\ln 2 \leq a < 0$. $\square$

The value of $\frac{2^a}{1+2^a}$ increases with $a < 0$ and is always less than $1/2$. Thus $X = 0$ is a non-empty set. From Eq. (5.3) and the expressions of v^2 and ω^2 in the variable v , we have

$$X = 0 \iff \frac{v^2}{\omega^2} = \frac{1}{2} \iff m = \frac{2(2-v)^a + v - 2}{v}.$$

Notice the expression after the last equivalence defines an actual mass only at the point where it intersects the graph of m given as a function of a and v by expression (2.7).

Lemma 5.7 Consider the function $n(a, v) = \frac{2(2-v)^a + v - 2}{v}$, $0 < v \leq 1$ defined on $\mathbb{R}^* \times (0, 1]$. For each a fixed, the function $n_a(v) = n(a, v)$ is increasing, and for each v fixed, the function $n_v(a) = n(a, v)$ is increasing. For every $a < 0$, $n_a(v)$ has a single zero at $v = 2 - 2^{\frac{1}{1-a}}$ and satisfies

$$n_a(1) = 1, \quad \text{and} \quad \lim_{v \rightarrow 0^+} n_a(v) = -\infty.$$

Proof The derivative of $n_a(v)$ is

$$\frac{1}{v^2} \{-2av(2-v)^{a-1} + 2[1 - (2-v)^a]\},$$

which is positive for all $a < 0$. The remaining claims are of straightforward verification. $\square$

Lemma 5.7 and some simple additional work imply that the graphs of $n_a(v)$ form a pencil of curves with a focal point at $(v, m) = (1, 1)$. The following proposition shows that the rhombus relative equilibria for which $0 < v \leq 1$, $0 < m \leq 1$ and $X = 0$ form a graph over the v -axis.

Proposition 5.8 *As a increases from $-\infty$ to 0, the graphs of $m_a(v)$ and $n_a(v)$ over $0 < v < 1$ intersect along a curve γ which is a graph with endpoints at the square $\mathcal{Q} = (0, 1)$ and the point $P = (1, 1/3)$.*

Proof Consider the functions

$$m_a(v) = \frac{1 - (2-v)^a}{1 - (2+v)^a}, \quad n_v(a) = \frac{2(2-v)^a + v - 2}{v} = n_a(v),$$

with $a < 0$ and $0 < v < 1$; the indices indicate a fixed variable. The limiting case $a = 0$ corresponds to $m_0(v) = \frac{\ln(2-v)}{\ln(2+v)}$ and $n_0(v) = 1$, so the intersection happens at $\mathcal{Q}$. As we decrease a , $n_v(a)$ decreases continuously from $m = n_0(v) = 1$ at $a = 0$ to $m = n_{-\infty}(v) = 1 - 2/v < 0$ at $a = -\infty$. As noted above, the graphs of $n_a(v)$, $a < 0$, have a focal point at $(v, m) = (1, 1)$. Since, as a varies from $-\infty$ to 0, the intersections of the graphs of $m_a(v)$ with the line $v = v_0$ form a segment from $m_0(v_0)$ to 1, there is a unique a_0 for which $(a_0, v_0, m_{a_0}(v_0)) = (a_0, v_0, n_{a_0}(v_0))$. Finally, we have that

$$m_a(v) = n_a(v) \iff (2-v)^a = \frac{2 + 2(4-v^2)^a + (2+v)^{a+1}}{2+v},$$

hence, along γ , we have the double limit

$$\lim_{\substack{a \rightarrow -\infty \\ v \rightarrow 1}} m = \lim_{\substack{a \rightarrow -\infty \\ v \rightarrow 1}} \frac{1 - (2-v)^a}{1 - (2+v)^a} = \frac{1}{3}.$$

$\square$

We call the expression

$$d^{(\tau, \alpha)} = \frac{\Delta^{(\tau, \alpha)}}{16\omega^4}$$

the *normalized discriminant*. Note that $16\omega^4$ has values 4^{3+2a} and $4^{3+a}(1+2^a)^2$ at the focal point $\mathcal{F} = (1, 0)$ and at the focal point $\mathcal{Q} = (0, 1)$, respectively.

Theorem 5.9 *The following optimal inequalities hold*

$$1 + a \leq d^{(\tau, \alpha)} \leq \begin{cases} d_0^{(\tau, \alpha)}, & \text{if } -\ln 3 / \ln 2 \leq a < 0, \\ d_{\diamond}^{(\tau, \alpha)}, & \text{if } a < -\ln 3 / \ln 2, \end{cases}$$

where $d_0^{(\tau, \alpha)}$ is the normalized discriminant of the oblate rhombus $\mathcal{F}$ and $d_{\diamond}^{(\tau, \alpha)}$ is the normalized discriminant of the square $\mathcal{Q}$.

Proof The theorem is a direct consequence of formula (5.2), identity (5.4), Proposition 5.6, the expression of $\Delta_{\diamond}^{(\tau, \alpha)}$ in the second paragraph of Sect. 4 and the expression

$$\Delta_0^{(\tau, \alpha)} = 4^{3+2a}(1+a+1/16a^2).$$

Optimality follows from Proposition 5.6 and the fact $\{X = 0\} \neq \emptyset$ which follows from Proposition 5.8. $\square$

Theorem 5.9 provides a reasonably complete description of the behavior of $d^{(\tau, \alpha)}$, and has important implications regarding linear stability. Firstly consider the straight line ℓ given by $d = d^{(\tau, \alpha)} \circ \gamma = 1 + a$, the curve $\pi_{\diamond}$ of equation $d = 1 + a + (2^a - 1)^2/[4(2^a + 1)^2]a^2$, which closely resembles a parabola, and the parabola π_0 defined by $d = 1 + a + 1/16a^2$. If $-\ln 3/\ln 2 < a < 0$, as we move from the square $\mathcal{Q}$ to the oblate rhombus $\mathcal{F}$ along the graph of $m_a(v)$, the relative positions of ℓ , $\pi_{\diamond}$ and π_0 indicate that $d^{(\tau, \alpha)}$ decreases from $\pi_{\diamond}$ to ℓ , then increases from ℓ to π_0 , crossing $\pi_{\diamond}$ once along the way. The curves $\pi_{\diamond}$ and π_0 intersect transversally when $a = -\ln 3/\ln 2$, and for all $a < -\ln 3/\ln 2$, $\pi_{\diamond}$ lies above π_0 , while ℓ is far below both π_0 and $\pi_{\diamond}$. For such values of a , as we go from $\mathcal{Q}$ to $\mathcal{F}$ with a fixed, $d^{(\tau, \alpha)}$ decreases from $\pi_{\diamond}$ to ℓ , crossing π_0 once on its way down, touches ℓ , then returns to π_0 . The proof of Proposition 5.6 justifies the monotonicity of all such movements of $d^{(\tau, \alpha)}$.

Linear stability in the subspace $\Pi^{(\tau, \alpha)}$ depends on the zero-set of $\Delta^{(\tau, \alpha)}$, which, according to Theorem 5.9, depends on the location of the points where ℓ , $\pi_{\diamond}$ and π_0 cross the a -axis. The following approximations are useful

$$a_{\diamond}^{(1)} \approx -1.03, \quad a_0^{(1)} = 4(-2 + \sqrt{3}) \approx -1.07, \quad a_{\diamond}^{(2)} \approx -3.89, \quad a_0^{(2)} = \frac{16}{a_0^{(1)}} \approx -14.93.$$

Theorem 5.10 *All rhombus relative equilibria (rhombi, for simplicity) are linearly stable if $a > -1$ with respect to displacements in the plane $\Pi^{(\tau, \alpha)}$. At $a = -1$, all rhombi are linearly stable except for $(-1, s_{-1}, m_{-1})$ and $(-1, 1/s_{-1}, 1/m_{-1})$, where $s_{-1} = 1/7(9 - 4\sqrt{2})$ and $m_{-1} = 1/7(11 + 6\sqrt{2})$. Assume $(s, m) \in [1, 3] \times [0, 1]$. For $a_{\diamond}^{(1)} < a < -1$, all rhombi remain linearly stable except for those in a small neighborhood of the curve γ from Proposition 5.8, and at $a = a_{\diamond}^{(1)}$, the square $\mathcal{Q}$ becomes unstable. For $a_0^{(1)} < a \leq a_{\diamond}^{(1)}$, the neighborhood of γ formed by unstable rhombi stretches up to $\mathcal{Q}$ along the graphs of each $m_a(v)$, and it finally reaches $\mathcal{F}$ when $a = a_0^{(1)}$. On the closed interval $a_{\diamond}^{(1)} \leq a \leq a_{\diamond}^{(2)}$, all rhombi are unstable. For $a < a_{\diamond}^{(2)}$, the square and a neighborhood thereof become linearly stable, a neighborhood of γ remains unstable. As a crosses the value $a_0^{(2)}$ in the negative direction, $\mathcal{F}$ and a neighborhood thereof become linearly stable. The stability of $(s, m) \in [1/3, 1] \times [1, \infty]$ is the same as that of $(1/s, 1/m) \in [1, 3] \times [0, 1]$.*

Proof Set $a = -1$. We have that

$$\Delta^{(\tau, \alpha)} = \left[\frac{7 - 18s + 7s^2}{(1 - 3s)(1 + s)} \right]^2.$$

The roots of $\Delta^{(\tau, \alpha)}$ are $1/7(9 \mp 4\sqrt{2})$ and we have $m_{-1}(1/7(9 \mp 4\sqrt{2})) = 1/7(11 \pm 6\sqrt{2})$. The last claim is a direct consequence of Lemma 5.3. The remaining assertions follow from Propositions 5.2 and 5.8, and from Theorem 5.9. $\square$

6 Linear Stability under Displacements in $\Pi^{(\phi, \psi)}$

Consider the expressions

$$\begin{aligned}\lambda_1^{(\phi, \psi)} + \omega^2 &= 2 \{m4^a s^a + (1+s)^a - (1+m)(1+s)^{a-1}[1 + (1+2a)s]\}, \\ \lambda_2^{(\phi, \psi)} + \omega^2 &= 2 \{m4^a s^a + (1+s)^a - (1+m)(1+s)^{a-1}(1+2a+s)\}, \\ \Delta^{(\phi, \psi)} &= 16(1+m)(1+s)^a [2\omega^2(1+a) + (1+m)(s-1)^2(1+s)^{a-2}a^2].\end{aligned}$$

Lemma 6.1 *For all $a \in \mathbb{R}^*$, the rhombus relative equilibria verify*

$$(\lambda_1^{(\phi, \psi)} + \omega^2)(\lambda_2^{(\phi, \psi)} + \omega^2) > 0.$$

Proof At $\mathcal{F} = (3, 0)$ the above product is $3a^2 4^{2a} > 0$. We write the product as a function of v and a , using (2.6) and (2.7). The expression thus obtained is

$$\frac{16^a (2-v)^{-2a}}{w_+^2} [4w_+^2 w_-^2 + 8aw_+ w_- (w_+ + w_-) + a^2 (4-v^2)(w_+ + w_-)^2],$$

with $w_{\pm} = 1 - (2 \pm v)^a$ and $-1 < v < 1$. Factoring $w_+^2 w_-^2$ out of the square brackets and setting $w = a(w_+^{-1} + w_-^{-1})$, the above expression becomes

$$16^a (2-v)^{-2a} w_-^2 [4 + 8w + (4-v^2)w^2],$$

which is clearly positive if a , hence w , is positive.

Assume a is negative. From

$$4 + 8w + (4-v^2)w^2 > 4 + 8w + 3w^2 = 3(w+2) \left(w + \frac{2}{3}\right),$$

we have that $4 + 8w + (4-v^2)w^2$ is positive if $w < -2$. To see that indeed $w < -2$, consider the derivatives of $w_{\pm}^{-1}$ with respect to v on $0 \leq v < 1$,

$$\frac{\partial w_{\pm}^{-1}}{\partial v} = \pm a \frac{(2 \pm v)^{-1+a}}{[1 - (2 \pm v)^a]^2}.$$

It follows that, for each $a < 0$, the derivative $\partial w / \partial v$ is negative, hence w has to be less than $w|_{v=0} = 2a(1-2^a)^{-1}$, which is less than $-2/\ln 2$ for every $a < 0$. $\square$

Proposition 6.2 *If $a \geq -1$, every rhombus relative equilibrium is linearly stable with respect to displacements in the subspace $\Pi^{(\phi, \psi)}$, with the exception of the square if $a = -1$.*

Proof The statement is a direct consequence of Proposition 3.3, the expression for $\Delta^{(\phi, \psi)}$, which is positive if $a > -1$ and zero if $a = -1$ and $s = 1$, and Lemma 6.1. $\square$

Observe that, if $a = -1$,

$$\lambda_1^{(\phi, \psi)} + \omega^2 = 2 \frac{-3 - 2v + v^2 + v^3}{(v+1)(v-2)} > 0, \quad \text{for } -1 < v < 1.$$

This shows the need of examining $\Delta^{(\phi, \psi)}$ when $a < 0$. In view of Proposition 6.2, for the purpose of stability analysis we may assume $a < -1$.

6.1 Analysis of $\Delta(\phi, \psi)$

Let us write

$$\Delta^{(\phi, \psi)} = 32(1+m)(1+s)^a \omega^2 \left[1 + a + \left(\frac{\eta^2}{\omega^2} \right) a^2 \right],$$

where $\eta^2 = \frac{1}{2}(1+m)(1-s)^2(1+s)^{a-2}$.

Lemma 6.3 *The ratio η^2/ω^2 is invariant under the inversion $\iota : (a, s, m) \mapsto (a, 1/s, 1/m)$. Hence the signs of $\Delta^{(\phi, \psi)}$ at (a, s, m) and $(a, 1/s, 1/m)$ are the same.*

Proof Fix $a \in \mathbb{R}^*$. From Eqs. (2.6) and (2.7), it follows that

$$\frac{\eta^2}{\omega^2} = \frac{1}{16} v^2 h(v), \quad \text{with} \quad h(v) = \frac{2 - (2-v)^a - (2+v)^a}{1 - (4-v)^a}.$$

The above expression for η^2/ω^2 is an even function of v . □

In view of Lemma 6.3, we may assume that $0 \leq v \leq 1$. After a few manipulations, the derivative of the function $h(v)$ in the proof of the lemma can be expressed as

$$2a \frac{[(2-v)^a - (2+v)^a][1 - (4-v^2)^a] + v(4-v^2)^a (\cosh \Upsilon_+ + \cosh \Upsilon_- - 2)}{(4-v^2)[1 - (4-v^2)^a]^2},$$

where $\Upsilon_{\pm} = a \ln(2 \pm v)$. Notice $a < 0$ implies $h'(v)$ is negative for all $v \in [0, 1]$, hence $h(v)$ is a decreasing function and the maximum of $h(v)$ on $0 \leq v \leq 1$ is

$$\max h(v) = h(0) = \frac{2}{1 + 2^a}. \quad (6.1)$$

We study the growth of $v^2 h(v)$. We have that $[v^2 h(v)]' = v[2h(v) + v h'(v)]$, and the sum in the square brackets is given by

$$\frac{2}{(4-v^2)[1 - (4-v^2)^a]^2} (A + aB),$$

where

$$A = (4-v^2) [2 - (2-v)^a - (2+v)^a] [1 - (4-v^2)^a],$$

and

$$B = v\{[(2-v)^a - (2+v)^a][1 - (4-v)^a] + v(4-v^2)^a (\cosh \Upsilon_+ + \cosh \Upsilon_- - 2)\}.$$

Proposition 6.4 *For every $a < 0$, the expression A is positive and decreases in value as v increases from 0 to 1. For every $a \leq -1$, the expression B is nonnegative and increases in value as v increases from 0 to 1.*

Proof Fix $a < 0$ and observe that $(2-v)^a + (2+v)^a$ is increasing on $0 \leq v \leq 1$. As a consequence, A is the product of three decreasing functions of v and is therefore decreasing.

Next we consider the product $B_1 = [(2-v)^a - (2+v)^a][1 - (4-v)^a]$. After a few manipulations, the derivative of B_1 with respect to v is given by $a/(4-v^2)$ times

$$\{2[(2-v)^a + (2+v)^a][-1 + (4-v^2)^a] + v[(2-v)^a - (2+v)^a][-1 + 3(4-v^2)^a]\}.$$

For $0 \leq v \leq 1$, the maximum value of $-1 + 3(4 - v^2)^a$ is $-1 + 3^{a+1}$, which is nonpositive if $a \leq -1$, so the above expression is the sum of a negative term and a nonpositive term. Thus B_1 has positive derivative with respect to v for every fixed $a \leq -1$. The product $B_2 = v(4 - v^2)^a (\cosh \Upsilon_+ + \cosh \Upsilon_- - 2)$ also has positive derivative with respect to v for every fixed $a \leq -1$. Indeed, we have that

$$\frac{\partial(\cosh \Upsilon_+ + \cosh \Upsilon_- - 2)}{\partial v} = \frac{a}{2} [(2 + v)^{a-1} - (2 - v)^{a-1} + (2 - v)^{-a-1} - (2 + v)^{-a-1}],$$

and, considering the expression in the square brackets, the difference of the first two terms is always negative, while the difference of the third and fourth terms is nonpositive as long as $a \leq -1$. We conclude that $B_1 + B_2$ is an increasing function of v for every $a \leq -1$ fixed, and so is $B = v\{B_1 + B_2\}$. $\square$

Proposition 6.4 implies that, on the interval $0 \leq v \leq 1$, A decreases from $A|_{v=0} = 8(1 - 2^a)(1 + 2^a)$ to $A|_{v=1} = 3(1 - 3^a)^2$, whereas, if $a \leq -1$, B increases from $B|_{v=0} = 0$ to $B|_{v=1} = (-1 + 3^a)^2 + 3^a[\cosh(a \ln 3) - 1]$. We also have that

$$\min(A + aB) = (A + aB)|_{v=1} = \frac{3}{2}(1 - 3^a)^2(a + 2), \quad \text{if } a \leq -1. \quad (6.2)$$

Corollary 6.5 *For every $-2 \leq a \leq -1$, the ratio η^2/ω^2 has maximum value $1/16$. If $a < -2$, the ratio η^2/ω^2 has upper bound $\frac{1}{8(1+2^a)}$.*

Proof Recall that $\eta^2/\omega^2 = \frac{1}{16}v^2h(v)$, and that

$$[v^2h(v)]' = \frac{2v}{(4 - v^2)[1 - (4 - v^2)^a]^2}(A + aB).$$

From Eq. (6.2) and Proposition 6.4, we deduce that $[v^2h(v)]'$ is positive if $-2 \leq a \leq -1$. Thus η^2/ω^2 attains its maximum value at $v = 1$, and since $h(1) = 1$, the maximum value of η^2/ω^2 is $1/16$.

Suppose now that $a < -2$. Equation (6.2) implies that $A + aB$ changes sign as v increases from 0 to 1. From the behavior of expressions A and B described in Proposition 6.4, we infer that $v^2h(v)$ has a single critical point $0 < v_c(a) < 1$, and that $v_c(a)^2h(v_c(a))$ is the maximum value of $v^2h(v)$ on the interval $0 \leq v \leq 1$. From Eq. (6.1) we infer that

$$\max\left(\frac{\eta^2}{\omega^2}\right) = \frac{1}{16}v_c(a)^2h(v_c(a)) < \frac{1}{16} \cdot 1^2 \cdot \frac{2}{1 + 2^a} = \frac{1}{8(1 + 2^a)}.$$

$\square$

As in Sect. 5, consider the normalized discriminant

$$d^{(\phi, \psi)} = \frac{\Delta(\phi, \psi)}{32(1 + m)(1 + s)^a\omega^2}.$$

Theorem 6.6 *The following inequalities hold*

$$1 + a \leq d^{(\phi, \psi)} \leq \begin{cases} d_0^{(\phi, \psi)}, & \text{if } -2 \leq a \leq -1, \\ 1 + a + \frac{1}{8(1+2^a)}a^2, & \text{if } a < -2, \end{cases}$$

and all inequalities are optimal except $d^{(\phi, \psi)} \leq 1 + a + \frac{1}{8(1+2^a)}a^2$ if $a < -2$, which is strict.

Proof $d_0^{(\phi, \psi)}$ denotes the normalized discriminant at the focal point $\mathcal{F}$, which corresponds to $v = 1$, so $d_0^{(\phi, \psi)} = 1 + a + 1/16a^2$. For the inequalities, it suffices to use the identity $d^{(\phi, \psi)} = 1 + a + \left(\frac{\eta^2}{\omega^2}\right)a^2$ and apply Corollary 6.5. Notice η^2/ω^2 only vanishes at $\mathcal{Q}$. $\square$

Remark 6.7 We have that $d_0^{(\tau, \alpha)} = d_0^{(\phi, \psi)}$ are optimal upper bounds for the normalized discriminants in $\Pi^{(\tau, \alpha)}$ and $\Pi^{(\phi, \psi)}$ as long as $-\ln 3/\ln 2 \approx -1.58 \leq a \leq -1$. Compare with Theorem 5.10.

We use Theorem 6.6 describe the behavior of $d^{(\phi, \psi)}$. Consider the straight line $\ell_\diamond$ given by $d = d_\diamond^{(\phi, \psi)} = 1 + a$, the parabola π_0 given by $d = 1 + a + 1/16a^2$ and the graph γ_c of the function $d = 1 + a + \frac{1}{16}v_c(a)^2h(v_c(a))a^2$. For $-2 \leq a \leq -1$ fixed, as we move from the oblate rhombus $\mathcal{F}$ to the square $\mathcal{Q}$ along the graph of $m_a(v)$, $d^{(\phi, \psi)}$ is negative and decreases from its value on π_0 to its value on $\ell_\diamond$. If $-6.82 \leq a < -2$, $d^{(\phi, \psi)} < 1 + a + \frac{1}{8}a^2$, so $d^{(\phi, \psi)}$ remains negative, and if $a < a_0^{(2)} \approx -14.93$, $d^{(\phi, \psi)}$ assumes positive values near $\mathcal{F}$ and negative values near $\mathcal{Q}$. In general, for each $a < -2$ fixed, as we move from $\mathcal{F}$ to $\mathcal{Q}$, $d^{(\phi, \psi)}$ increases from π_0 to γ_c and then decreases to its (negative) minimum at $\ell_\diamond$.

Remark 6.8 Our numerical experiments suggest that $0.9 \leq v_c(a) \leq 1$ for every $a \leq -2$. If $a = -11$, which is near the point where γ_c crosses the a -axis in the positive d -direction, linearly stable rhombi appear at $v \approx 0.88$ and $m \approx 0.71$. The experiments seem to point out that, for decreasing values of $a < -10.89$, the zero of $d^{(\phi, \psi)}$ approaches $v = 0$, $m = 1$ (the square).

Recall the notation used in Theorem 5.10.

Theorem 6.9 Consider only displacements in the plane $\Pi^{(\phi, \psi)}$. All rhombus relative equilibria are linearly stable provided $a > -1$, and the square is the only unstable rhombus if $a = -1$. For every $a < -1$, the square $\mathcal{Q}$ and a neighborhood thereof are unstable. All rhombus relative equilibria are unstable if $-6.82 \leq a \leq a_0^{(1)} \approx -1.07$, and, for $a_0^{(1)} < a < -1$, all rhombi in a neighborhood of the oblate rhombus $\mathcal{F}$ are linearly stable. If $a < a_0^{(2)} \approx -14.93$, a neighborhood of $\mathcal{F}$ is linearly stable. The stability of $(s, m) \in [1/3, 1] \times [1, \infty]$ is the same as that of $(1/s, 1/m) \in [1, 3] \times [0, 1]$.

Numerical analysis of the critical curve γ_c indicates that, for $-10.89 \leq a < -6.82$, all rhombus relative equilibria are unstable if subject to displacements in $\Pi^{(\phi, \psi)}$.

7 Closing Remarks

We conclude the article with some illustrations of the behaviour of the normalized discriminants $d^{(\tau, \alpha)}$ and $d^{(\phi, \psi)}$. Figure 6 displays the graphs of $d^{(\tau, \alpha)}$ and $d^{(\phi, \psi)}$ for values of a representing the main possibilities for their positions relative to one another. When the graphs of both discriminants are above the v -axis, linear stability occurs, otherwise we have instability. If $a = -15$, for instance, there are two intervals of linear stability on the v -axis, one approximately bounded by 0.71 ($m \approx 0.98$) and 0.93 ($m \approx 0.63$), and the other bounded by $v \approx 0.99995$ ($m \approx 0.00075$) and 1 ($m = 0$). The fast changes in sta-

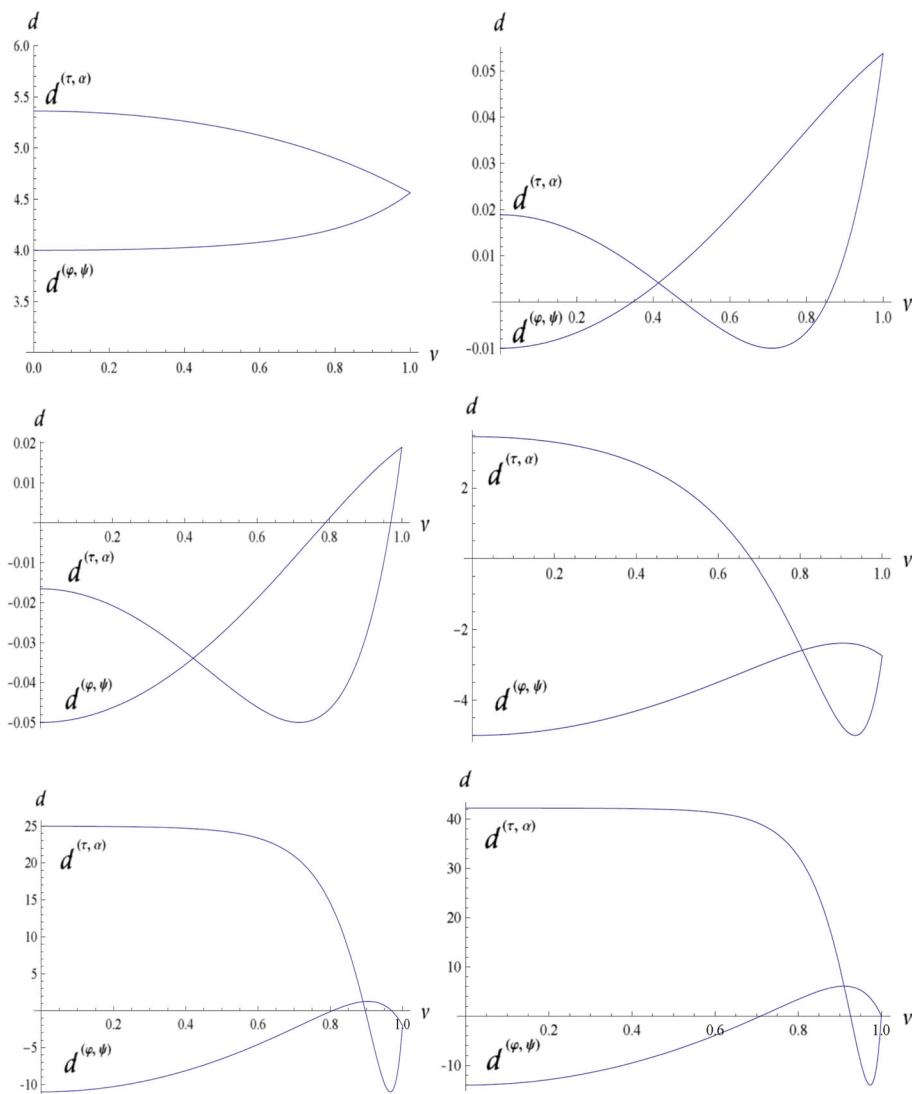


Fig. 6 The normalized discriminants for $a = 3, -1.01, -1.05, -6, -12$ and -15 (left to right, top to bottom)

bility that happen from $a = -1$ to $a \approx -1.07$ are depicted in the second frame of the first row and the frames of the second row (graphs for $a \approx -1.07$ and $a = -6$ behave similarly).

Concerning Moeckel's dominant mass conjecture, see the Introduction, Theorems 5.10 and 6.9 imply that the conjecture gives the correct stability result if $-6.82 \lesssim a \lesssim -1.07$, and, according to numerical evidence, the lower bound of this interval can be improved from $a = -6.82$ to approximately -10.89 . In contrast, Theorems 5.10 and 6.9 also imply that Moeckel's conjecture is false if $a > -1.07$ and $a \lesssim -14.93$, and numerical plots suggest the latter estimate can be improved to $a \lesssim -10.89$. Again we refer to Fig. 6 for illustrations of the changes in stability that occur when we vary the exponent a .

Appendix A: Diagonalization of $D\nabla U_a(\mathbf{x}^{(0)})$

In this appendix we explain how group representation theory was applied to simplify the linear stability analysis of a rhombus relative equilibrium. References are [5, 12]. The key basic fact is that the rhombus is a configuration with a nontrivial symmetry group Σ which can be viewed as the dihedral group D_2 generated by the reflections through the diagonals of the rhombus, or as Klein's four-group V , which is generated by the reflections through the perpendicular bisectors of opposite sides of the rhombus. As D_2 , Σ consists of rotations E and R about the midpoint of one of the diagonals by zero and 180 degrees, respectively, a reflection S across the same diagonal, and a reflection RS through the other diagonal. As explained in [5], it follows that the restriction B of the hessian matrices $D\nabla U_a(\mathbf{x}^{(0)})$ of the potentials U_a to the space of planar displacements of a rhombus relative equilibrium $\mathbf{x}^{(0)}$ commutes with the elements of a group of isomorphisms of that space. This observation, together with our knowledge of some of the eigenvectors of the hessian, can be used to fully diagonalize the matrices $M^{-1}B$, as shown by our calculations in Sect. 3.2.

Let G be a finite group. A (linear) representation of G on the (complex, real) vector space $\mathcal{V}$ is a map ρ from G into the group $GL(\mathcal{V})$ of isomorphisms of $\mathcal{V}$. A subrepresentation of ρ is a representation of G on $\mathcal{V}$ given by the restriction of ρ to a subspace $\mathcal{W}$ of $\mathcal{V}$ which is ρ -invariant, i.e., such that $\rho(g)(\mathcal{W}) \subseteq \mathcal{W}$ for all $g \in G$. A representation is said to be irreducible if its only subrepresentations are the ones corresponding to the trivial subspaces $\{0\}$ and $\mathcal{V}$. The table below contains a description of the irreducible representations of D_2 .

The entries of the column headed by E indicate that τ, α, ϕ, ψ are one-dimensional representations.

Every complex representation ρ of the finite group G can be written (non-uniquely) as the direct sum of subrepresentations isomorphic to the irreducible representations of G . The direct sum of all copies of a given irreducible representation ϱ of G present in ρ is the space $\mathcal{V}_\varrho$ of a subrepresentation of ρ called the ϱ -isotypic component of ρ . For each irreducible representation $\varrho_1, \dots, \varrho_t$ of the finite group G and representation $\rho : G \rightarrow GL(\mathcal{V})$, representation theory provides a family of projection operators p_j whose images are precisely the isotypic components $\mathcal{V}_{\varrho_j}$. Using the projections p_k , we obtain the isotypic decomposition of $\mathcal{V}$, namely

$$\mathcal{V} = \mathcal{V}_{\varrho_1} \oplus \dots \oplus \mathcal{V}_{\varrho_t}.$$

In the case of D_2 , we have that

$$\begin{aligned} p_\tau &= \frac{1}{4}[\rho(E) + \rho(R) + \rho(S) + \rho(RS)], & p_\alpha &= \frac{1}{4}[\rho(E) + \rho(R) - \rho(S) - \rho(RS)], \\ p_\phi &= \frac{1}{4}[\rho(E) - \rho(R) + \rho(S) - \rho(RS)], & p_\psi &= \frac{1}{4}[\rho(E) - \rho(R) - \rho(S) + \rho(RS)], \end{aligned}$$

where the coefficients of the linear combinations in the square brackets on the right-hand sides of the above expressions are precisely the entries of each row of Table 1.

We will apply the projections $p_\tau, p_\alpha, p_\phi, p_\psi$ associated to the representation of D_2 on the space of planar displacements of the rhombus (identified with $\mathbb{R}^8$) defined by

$$\sigma(g)(\delta) = (g\delta_{g^{-1}(1)}, g\delta_{g^{-1}(2)}, g\delta_{g^{-1}(3)}, g\delta_{g^{-1}(4)}),$$

for all $g \in D_2$, $\delta = (\delta_1, \delta_2, \delta_3, \delta_4) \in \mathbb{R}^8$. In our notation, each δ_j is a vector in a copy of $\mathbb{R}^2$ whose origin is at the j th vertex of the rhombus. In order to understand the definition of σ , notice that the action of each $g \in D_2$ on the set of vertices of the rhombus produces a permutation of the vertices, i.e., g can be interpreted as a permutation of the indices 1, 2, 3, 4

Table 1 The irreducible representations of the group D_2

	E	R	S	RS
τ	1	1	1	1
α	1	1	-1	-1
ϕ	1	-1	1	-1
ψ	1	-1	-1	1

of the vertices. As an element of $\mathbb{R}^2$, each displacement δ_j can be rotated or reflected according to $g \in D_2$. Thus the definition of σ implies that each $\sigma(g)$ corresponds to applying g as a rigid motion of the rhombus together with its displacement δ . The isotypic decomposition of the space of displacements is the direct sum of the images (isotypic components) $\mathcal{V}_\tau, \mathcal{V}_\alpha, \mathcal{V}_\phi, \mathcal{V}_\psi$ of $p_\tau, p_\alpha, p_\phi, p_\psi$. A simple application of character theory shows that the dimensions of the isotypic components are all equal to two, which is the common multiplicity of τ, α, ϕ, ψ in σ .

Before moving on, we introduce an Euclidean structure in the space of displacements of a rhombus relative equilibrium $\mathbf{x}^{(0)}$ using the masses, namely

$$\langle (\delta_1, \delta_2, \delta_3, \delta_4), (\varepsilon_1, \varepsilon_2, \varepsilon_3, \varepsilon_4) \rangle_M = m\delta_1 \cdot \varepsilon_1 + \delta_2 \cdot \varepsilon_2 + m\delta_3 \cdot \varepsilon_3 + \delta_4 \cdot \varepsilon_4,$$

where the $\cdot$ represents the usual dot product in $\mathbb{R}^2$. It is not hard to verify that the projections $p_\tau, p_\alpha, p_\phi, p_\psi$ are orthogonal with respect to $\langle \cdot, \cdot \rangle_M$, or M -orthogonal for short. As a consequence, the isotypic decomposition

$$\mathbb{R}^8 = \mathcal{V}_\tau \oplus \mathcal{V}_\alpha \oplus \mathcal{V}_\phi \oplus \mathcal{V}_\psi$$

is an M -orthogonal decomposition of the space of displacements of $\mathbf{x}^{(0)}$.

Let us firstly determine $\mathcal{V}_\tau$. If we apply p_τ to the displacement $v_1 = (r, 0, 0, 1, 0, 0, 0, 0)$, we obtain $\delta_\tau = \mathbf{x}^{(0)}$ viewed as a displacement. By applying p_τ to $w_1 = (1/(mr), 0, 0, -1, 0, 0, 0, 0)$, we obtain a displacement ε_τ which, together with δ_τ , forms an M -orthogonal basis of $\mathcal{V}_\tau$. Figure 3 of subsection 3.2 depicts the images of v_1 and w_1 under p_τ .

In order to determine $\mathcal{V}_\alpha$, we apply a rotation of 90 degrees to the copies of $\mathbb{R}^2$ at each vertex of the rhombus. As in Sect. 3.2, denote by J the corresponding linear map. We have that

$$J \circ \sigma(g) = \sigma(g) \circ J, \text{ if } g = E, R, \text{ and } J \circ \sigma(g) = -\sigma(g) \circ J, \text{ if } g = S, RS.$$

These simple relations imply the following relations among the projections

$$J \circ p_\tau = p_\alpha \circ J, \text{ and } J \circ p_\phi = p_\psi \circ J.$$

Thus an M -orthogonal basis of $\mathcal{V}_\alpha$ can be produced simply by applying J to the displacements forming the M -orthogonal basis of $\mathcal{V}_\tau$ determined in the previous paragraph. We illustrated the basis of $\mathcal{V}_\alpha$ formed by $\delta_\alpha = J\delta_\tau$ and $\varepsilon_\alpha = J\varepsilon_\tau$ in Fig. 4, Sect. 3.2.

Next we apply the same procedure in order to obtain M -orthogonal bases for $\mathcal{V}_\phi$ and $\mathcal{V}_\psi$, with v_1, w_1 replaced with $v_2 = (1, 0, 1, 0, 0, 0, 0, 0)$ and $w_2 = (1/m, 0, -1, 0, 0, 0, 0, 0)$. By applying p_ϕ to v_2 and w_2 , we obtain the displacements δ_ϕ and ε_ϕ on the first two columns of Fig. 5, Sect. 3.2. Notice δ_ϕ and $\delta_\psi = J\delta_\phi$ correspond respectively to horizontal and vertical translations of the rhombus.

We collect the displacements determined in the previous paragraphs and reorder them so that J keeps its block-diagonal matrix form. The basis of the space of planar displacements of the rhombus thus obtained consists of the displacements

$$\delta_\tau = (r, 0, 0, 1, -r, 0, 0, -1), \quad (\text{A.1})$$

$$\delta_\alpha = (0, r, -1, 0, 0, -r, 1, 0), \quad (\text{A.2})$$

$$\varepsilon_\tau = (1/(mr), 0, 0, -1, -1/(mr), 0, 0, 1), \quad (\text{A.3})$$

$$\varepsilon_\alpha = (0, 1/(mr), 1, 0, 0, -1/(mr), -1, 0), \quad (\text{A.4})$$

$$\delta_\phi = (1, 0, 1, 0, 1, 0, 1, 0), \quad (\text{A.5})$$

$$\delta_\psi = (0, 1, 0, 1, 0, 1, 0, 1), \quad (\text{A.6})$$

$$\varepsilon_\phi = (1/m, 0, -1, 0, 1/m, 0, -1, 0), \quad (\text{A.7})$$

$$\varepsilon_\psi = (0, 1/m, 0, -1, 0, 1/m, 0, -1). \quad (\text{A.8})$$

It can be deduced from theory [5] that vectors (A.1)–(A.8) must form a basis of eigenvectors of B and $M^{-1}B$, where M is the matrix that defines the inner product $\langle \cdot, \cdot \rangle_M$. This basis can be found in previous articles, see for instance [10], however the deeper connection with the symmetry of the rhombus has not been clarified as far as we know.

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